Week 5 Material - Implicit Derivatives, Related Rates and Optimization

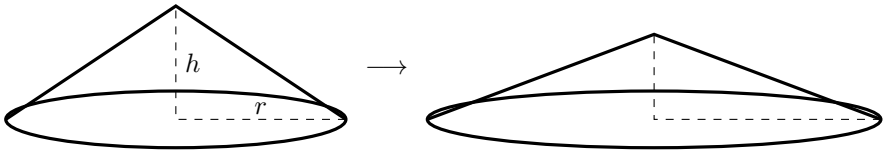
19. [1] Find the slope of the tangent line to the curve $y^2 + xy - y^3 = 2$ at the point (3,2).

20. [2] A wedding photographer is using an aerial drone (i.e. a small helicopter) to film a wedding. She keeps the drone at a steady height of 8 m above the ground, and moving horizontally at 0.5 m/s.

    As the drone moves, the photographer has to rotate the camera attached to the bottom of the drone to keep the bride and groom in view. When the drone is moving towards the bride and groom, but still 30 m away *horizontally* from them, at what rate would the camera need to be rotated/turned to keep the camera aimed at the bride and groom? Measure the angle as the angle below horizontal, and include units in your answer.

    We recommend drawing a diagram to start. You must define any variables you introduce, either through a diagram or in words.

21. [3] A cone made of sand is collapsing under its own weight, causing the height of the cone to decrease. None of the sand is blowing away so **the volume of the cone remains constant**. As the height decreases the base spreads out. The current height of the cone is 2 m and its base has a radius of 3 m. If the height of the cone is decreasing at a rate of 0.1 m/s then how fast is the circumference of the base increasing?



    The volume of a cone is given by $V = \frac{1}{3}\pi r^2 h$ where r is the radius of the circular base of the cone and h is its height.

    Note: Optimization problems were re-included in APSC 171 in 2020, so older tests do not typically include problems of this type. They **are** testable for this year! See the Practice Problems for more examples.

22. A 15 cm x 8 cm rectangular piece of cardboard is to be made into an open-topped box by cutting from each corner a square of side length x, and then folding up the remaining edges; this will make a box of height x.



    (a) Write out a formula for the volume of the box.

    (b) What side length of square should be cut away to maximize the volume of the box? Compute the resulting volume as well, and include units in your answer.

23. A rectangle is inscribed with its base on the x axis and its upper corners on the parabola $y = 20 - 4x^2$. What are the dimensions of such a rectangle with the greatest possible area? Compute the final rectangle area associated with those dimensions.

[1] Source: previous APSC 171 Test 1
[2] Source: previous APSC 171 Test 2
[3] Source: previous APSC 171 Test 1